Prabhavit Gupta

AI Platform Engineer | Agent Runtimes, LangGraph, MCP Servers & LLMOps

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Executive Summary

- **Core Agent Platform & Runtime Engineering:** AI Engineer with 3 years commercial software experience specializing in agent orchestration runtimes, shared developer SDKs, and state/memory management. Proven track record architecting modular agent systems with LangGraph, Anthropic/OpenAI SDKs, and robust async Python microservices.
- **Tool & Connector Layers via Model Context Protocol (MCP):** Extensive hands-on experience authoring and maintaining custom MCP (Model Context Protocol) servers that safely expose structured enterprise databases, internal APIs, and search infrastructure to autonomous LLM agents with defensive JSON validation.
- **LLMOps, Distributed Observability & Evals:** Deep expertise implementing enterprise-grade telemetry with Langfuse, semantic caching layers with Redis, and automated CI/CD evaluation harnesses using RAGAS and custom LLM-as-a-judge pipelines to guarantee prompt stability and sub-second retrieval latency.

Technical Skills & Core Competencies

- **Agent Architecture & Orchestration:** LangGraph, LangChain, Model Context Protocol (MCP) Servers, Multi-Agent Coordination, Memory Management, Tool Calling, State Machines, Context Pruning.
- **Core Engineering & Languages:** Python (AsyncIO, FastAPI, Pydantic, PyTest), TypeScript / Node.js, REST APIs, WebSockets, Docker, Kubernetes basics, Git, GitHub Actions CI/CD.
- **Databases & Vector Retrieval:** PostgreSQL (pgvector, HNSW indexes), Qdrant, Redis (Semantic Caching & Task Queues), Hybrid Retrieval (Vector + BM25 Lexical Re-ranking).
- **LLMOps, Security & Observability:** Langfuse Distributed Tracing, RAGAS Eval Harnesses, Prompt Injection Mitigation, PII Redaction, Token Cost & Latency Profiling.
- **Engineering Standards:** Modular SDK Design, Secure-by-Default Architecture, High-Leverage Internal Tooling, Extreme Ownership.

Professional Experience

AI Engineer & Platform Consultant • *Independent / AI Product Engineering Consultancy*
2024 – Present • Venice, IT (Remote Europe)

- Engineered reusable internal agent runtimes and orchestration layers utilizing LangGraph and Python AsyncIO, enabling rapid deployment of multi-step autonomous workflows.
- Implemented and deployed Model Context Protocol (MCP) servers exposing internal analytical databases and web intelligence endpoints to LLMs with strict input validation and access controls.
- Established automated evaluation pipelines using RAGAS and regression test suites in GitHub Actions, benchmarking ground truth relevance, faithfulness, and latency across prompt iterations.
- Integrated Langfuse distributed tracing across all agent execution paths, giving development teams real-time visibility into token expenditures, tool invocation latency, and failure fallbacks.
- Architected high-throughput hybrid RAG systems over relational and unstructured data sources, combining PostgreSQL pgvector embeddings with lexical keyword re-ranking.
- Designed semantic caching middleware with Redis, eliminating redundant upstream LLM calls by 35% while preserving freshness guarantees for dynamic analytical data.

Full-Stack AI Software Engineer • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Built resilient asynchronous backend microservices in Python (FastAPI) and TypeScript powering automated data extraction and semantic document classification.
- Integrated vector indexing and semantic similarity matching over high-volume transactional records using PostgreSQL and Redis caching.
- Formulated end-to-end integration test suites with PyTest and Dockerized staging environments, achieving 90%+ test coverage across core integration layers.
- Collaborated with product and engineering teams to transform exploratory AI prototypes into production-grade, observable microservices.

Key Engineering Projects

- **Enterprise Agent Runtime & MCP Connector Gateway (Python + LangGraph + MCP):** Created a modular agent execution runtime featuring shared state management, dynamic tool routing via MCP servers, and comprehensive Langfuse trace capture for enterprise automation.
- **Continuous LLM Eval & Regression Gate (FastAPI + RAGAS + PyTest):** Implemented an automated

CI/CD eval suite that executes synthetic test cases on prompt updates, verifying contextual recall, adherence, and latency constraints prior to production releases.

Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy & Economics** – Ca' Foscari University of Venice (2018 – 2022)
  Deep background in formal logic, statistical reasoning, decision architecture, and quantitative methodology.
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Czech (Willing to learn).
- **Work Authorization & Legal Status:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy) – Unrestricted legal right to live and work across the European Union (including the Czech Republic) without visa sponsorship. Immediate availability.